Islam Abd-Elhady

AI/ML Engineer | LLM Engineer | MLOps Engineer
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PROFESSIONAL SUMMARY

AI and Machine Learning Engineer with over two years of experience building production-grade large language model systems, retrieval augmented generation pipelines, and multi-agent architectures serving thousands of concurrent users. Specialize in LLMOps, MLOps, GPU optimization, and scalable microservices on AWS, Azure, and Kubernetes. Delivered platforms that reduced design cycles by 90 percent, lowered GPU memory usage by 40 percent, and sustained high throughput inference across distributed workloads. Proficient in Python, PyTorch, LangChain, LangGraph, FastAPI, Docker, and continuous integration and delivery.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, Java, C Sharp, Go, Bash, SQL
- **Large Language Models:** LLM fine-tuning, LoRA, PEFT, QLoRA, prompt engineering, agent orchestration, function calling, token optimization, guardrails, model evaluation
- **LLM Frameworks:** LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, Diffusers, LiteLLM, OpenAI, Anthropic, Google Gemini
- **Generative AI:** Stable Diffusion, SDXL, Playground v2.5, Real-ESRGAN, SwinIR, CLIP, DINOv2, SAM, faster-whisper, retrieval augmented generation
- **Machine Learning and Deep Learning:** PyTorch, TensorFlow, scikit-learn, convolutional neural networks, transformers, computer vision, natural language processing
- **Vector Databases and Retrieval:** FAISS, Qdrant, Chroma, Pinecone, hybrid retrieval, re-ranking, embeddings
- **MLOps and LLMOps:** model deployment, inference optimization, model versioning, experiment tracking, A/B testing, continuous integration and delivery
- **Observability and Optimization:** structured logging, metrics, distributed tracing, GPU optimization, xFormers, ONNX, TensorRT, quantization, distributed training
- **Cloud Platforms:** AWS, EC2, S3, IAM, GPU instances g5.4xlarge and g5.8xlarge, Azure virtual machines, Azure Container Apps, Azure Blob Storage, SaladCloud
- **Infrastructure and DevOps:** Docker, Kubernetes, Helm, systemd, GitHub Actions, continuous integration and delivery pipelines
- **Backend and System Design:** FastAPI, WebSocket, REST API design, asynchronous API, microservices architecture, distributed systems, scalable systems
- **Backend Services:** Redis, Celery, RQ, message queues, caching, load balancing, rate limiting
- **Data Engineering:** Pandas, NumPy, ETL pipelines, data pipelines, data validation, web scraping, Apify, Selenium, BeautifulSoup
- **Security and Reliability:** sandboxed execution, AST validation, input sanitization, secrets management, high availability, fault tolerance
- **Developer Tools:** Git, GitHub Actions, Linux, Ubuntu, Bash scripting, unit testing, integration testing

PROFESSIONAL EXPERIENCE

AI and DevOps Engineer Nov 2025 – Present
Zedny INC, Cairo, Egypt (On-site)

- Architected production-grade AI platforms on microservices, delivering scalable large language model inference services across distributed GPU clusters serving over 10,000 daily requests with 99.9 percent uptime.
- Designed end-to-end MLOps and LLMOps pipelines covering data ingestion, training, evaluation, serving, and model versioning, reducing deployment cycle time by 60 percent.
- Orchestrated GPU workloads with Docker and Kubernetes through autoscaling and resource quotas, increasing throughput by 45 percent while cutting cloud costs by 30 percent.
- Built continuous integration and delivery pipelines with GitHub Actions for automated testing, container builds, and

zero-downtime deployments, shipping over 50 releases per month.
- Implemented observability stacks with structured logging, metrics, and distributed tracing, reducing mean time to detect production issues by 70 percent.
- Optimized inference cost and latency through request batching, caching, and model quantization, lowering p95 latency by 35 percent and cloud spend by 25 percent.
- Defined API contracts, system boundaries, and service level objectives with product teams, translating AI requirements into scalable production services.
- Hardened multi-tenant AI deployments with secrets management, rate limiting, and failover strategies.

AI and Machine Learning Engineer Sep 2024 – Oct 2025
VEEM Solutions (Shrwd.ai Platform), Jeddah, Saudi Arabia (Remote)

- Developed a large-scale multi-agent large language model system with dynamic tool orchestration and a retrieval augmented generation knowledge layer, enabling autonomous task execution across more than five business domains.
- Engineered multilingual retrieval pipelines using FAISS vector search, MiniLM embeddings, hybrid retrieval, and re-ranking, improving answer relevance by 40 percent and reducing hallucinations by 50 percent.
- Applied prompt engineering, model evaluation, and guardrails with schema validation to stabilize agent behavior across production traffic handling over 1,000 daily conversations.
- Architected the Shrwd.ai brand intelligence platform integrating large-scale web crawlers, SEO analytics, and interactive dashboards for real-time tracking of more than 100 brands.
- Created real-time FastAPI and WebSocket services with Redis caching and Celery workers, sustaining over 500 concurrent large language model inference sessions under production load.
- Deployed services on Azure virtual machines and Container Apps with Docker and continuous integration and delivery, achieving zero-downtime releases across staging and production.
- Instrumented pipelines with logging, metrics, and alerting to track retrieval quality, token usage, and latency, enabling data-driven iteration on prompts and models.

AI Engineer (Part-time) Nov 2024 – Dec 2025
DigiTee.io, Cayman Islands (Remote)

- Led development of NFT Wear AI, fine-tuning Stable Diffusion with LoRA to auto-generate apparel mockups, compressing design cycles from over four hours to under five minutes, a 95 percent reduction.
- Created generative pipelines for the Mintora platform using Playground v2.5 and SDXL, integrating conversational user experience, rembg background removal, and Real-ESRGAN and SwinIR upscaling.
- Architected and shipped Audira, a multi-agent voice platform with memory, faster-whisper speech to text, text to speech, and tool orchestration, enabling real-time multilingual onboarding.
- Deployed and scaled GPU inference on AWS g5.4xlarge and g5.8xlarge instances with queue-based job distribution and load balancing, sustaining over 200 concurrent inference requests.
- Reduced GPU out-of-memory incidents by 80 percent and improved throughput by 40 percent through xFormers, gradient checkpointing, memory efficient attention, and mixed precision inference.
- Owned deployment, monitoring, and incident response across environments, including log aggregation and GPU utilization tracking for production inference workloads.

Freelance AI and Machine Learning Engineer 2024 – 2026
Independent Consulting, Remote

- Delivered over ten retrieval augmented generation systems for international clients, integrating FAISS, Hugging Face embeddings, FastAPI, and WebSocket APIs for contextual question answering at scale.
- Engineered custom multi-agent architectures with LangChain and LangGraph, automating complex business workflows through external tool and API integration.
- Created real-time voice assistants combining faster-whisper speech to text and multilingual text to speech pipelines, achieving sub-second response latency.
- Deployed AI services across Azure, AWS, and SaladCloud, owning model selection, fine-tuning, infrastructure provisioning, and post-deployment monitoring.
- Advised clients on architecture tradeoffs, cost optimization, and evaluation strategies, translating ambiguous requirements into production-ready AI solutions.

AI Instructor Oct 2024 – Apr 2025
USAID, Ahmed Diefalla International Applied Technology School, Cairo, Egypt

- Delivered structured curricula in software engineering, Python, AI, and robotics to over 100 students through project-

based learning.
- Guided end-to-end capstone projects covering problem framing, implementation, and deployment, strengthening technical and analytical skills.

Programming Content Creator (Part-time) Mar 2024 – Mar 2025
Tweet Kod, Karabuk, Turkey (Remote)

- Authored Python programming curricula for youth learners, producing over 50 hands-on projects and assessment modules.
- Mentored students through graduation projects, improving coding proficiency and engineering reasoning.

Programming and AI Instructor (Part-time) Jan 2024 – Dec 2024
Future Kids, Codemy, Code and Barmja, Cairo, Egypt (Remote)

- Taught Scratch, Python, Java, HTML, AI fundamentals, and Pygame across diverse age groups.
- Designed tailored lesson plans fostering creativity, critical thinking, and programming confidence for over 200 learners.

SELECTED PROJECTS

DataOps LLM Engine GitHub Link

- Built a large language model powered data operations engine enabling natural language interaction with Excel, CSV, and pandas DataFrames by translating user intent into safe executable Python code.
- Engineered a multi-provider abstraction layer via LiteLLM supporting OpenAI, Anthropic, and Google Gemini with unified prompting, failover, and cost-aware routing, reducing provider lock-in risk.
- Designed a seven-layer security architecture featuring AST validation, import whitelisting, sandboxed execution, resource limits, timeout enforcement, output sanitization, and audit logging.
- Delivered a FastAPI REST interface with asynchronous execution and schema validation, supporting over 100 requests per second with structured JSON responses.
- Optimized for developer usability and scalability, exposing a clean software development kit and high throughput API with separation between reasoning, validation, and execution layers.

Eyes Mate: AI Assistant for the Visually Impaired GitHub Link

- Developed a production-style assistive AI system combining object recognition, distance estimation, optical character recognition, currency recognition, and face recognition with real-time voice feedback.
- Integrated speech and computer vision pipelines into a unified low-latency experience, achieving response times under two seconds on mobile hardware.

Brain Tumor Detection API GitHub Link

- Built a convolutional neural network pipeline for brain tumor classification from MRI scans, reaching over 95 percent validation accuracy, packaged as a production-ready REST API with versioned endpoints.
- Designed structured request and response schemas with error handling for safe integration into downstream medical applications.

Generative AI Models GitHub Link

- Implemented and benchmarked transformer and diffusion-based architectures for text-to-image and image-to-text tasks across multiple datasets.
- Explored fine-tuning, evaluation, and prompt strategies for creative and practical generative AI applications.

SmartAI Customer Support GitHub Link

- Designed a retrieval augmented generation powered customer support assistant with multilingual grounding, improving answer accuracy by 45 percent over baseline large language model responses.
- Integrated retrieval, re-ranking, and generation into a unified inference service with structured logging and evaluation hooks.

Deep CUDA: GPU Acceleration from Scratch GitHub Link

- Implemented a full convolutional neural network in raw CUDA C++ for MNIST classification with custom kernels for convolution, matrix multiplication, and backpropagation without high-level frameworks.

- Optimized GPU performance through shared memory, thread and block scheduling, and parallel reduction, achieving a tenfold speedup over the CPU baseline.

EDUCATION

Master of Science in Data Science (Preparatory Year) 2026 – Present
Faculty of Computers and Artificial Intelligence, Helwan University, Cairo, Egypt
Focus areas: machine learning, statistical analysis, applied data science.

Bachelor of Science in Computer Science 2020 – 2024
Faculty of Computers and Information, Assiut University, Assiut, Egypt
Relevant coursework: machine learning, deep learning, computer vision, parallel computing.

CERTIFICATIONS

- **LLMOps and AIOps Bootcamp with Eight End-to-End Projects**, Udemy, KRISHAI Technologies Mar 2026
  Certificate Link
- **Generative AI with Large Language Models**, Coursera Apr 2025
  Certificate Link
- **Advanced Learning Algorithms**, DeepLearning.AI Sep 2023
  Certificate Link
- **Deep Learning with PyTorch: Image Segmentation**, Coursera Sep 2023
  Certificate Link
- **Empedia Competitive Programming Competition**, Empedia Aug 2023
  Certificate Link
- **Mathematics for Computer Science**, BIO CODE, Assiut University Aug 2023
  Certificate Link
- **Supervised Machine Learning: Regression and Classification**, DeepLearning.AI Mar 2023
  Certificate Link
- **Python Programming Basics**, MaharaTech Feb 2023
  Certificate Link
- **Database Fundamentals**, MaharaTech Feb 2023
  Certificate Link

PUBLICATIONS AND RESEARCH

- **AI Traffic Violation in Smart Cities and Flying Vehicles.** Research on AI-driven real-time monitoring, detection, and enforcement systems for intelligent transportation and urban mobility.

LANGUAGES

Arabic (Native) | English (Professional)